LOGISTIC REGRESSION PROJECT 2

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Introduction

In this report, concentrations of plasma beta-carotene levels in human individuals will be studied in relation to different explanatory variables, using logistic regression. The explanatory variables are given in Table 1. To study different relationships using different logistic models, data from a study in which 315 individuals participated will be used.

Variable	**Description**
age	(In years)
sex	(Male or female)
smokstat (Smoking)	Never/former/current smoker
bmi	Body mass index, weight/height^2
vituse (Vitamin use)	Vitamin use (regularly/irregularly/no)
calories	Daily calories consumed (MJ)
fat	Daily fat consumed (g)
fiber	Daily fiber consumed (g)
alcohol	Number of alcoholic drink consumed weekly
cholesterol	Daily consumed cholesterol (mg)
betadiet (Beta-carotene)	Daily dietary beta-carotene consumed (mg)

Table 1. Explanatory variables with short descriptions.

1. Low plasma beta-carotene and vitamin use

In this first section of the report, we will study the relationship between beta-carotene levels and vitamin use among participants in the study. To study the relationship between beta-carotene levels and different covariates using logistic regression, a cut-off value needs to be chosen. The cut-off value indicates where we draw the line between *low* and *not low* (referred to as "high") beta-carotene levels. In the logistic regression models that are soon to be introduced, the binary response variable is a "1" if beta-carotene levels are below the cut-off value, and a "0" otherwise.

The cutoff value is decided as 0,42 μmol/l (micromoles per liter), based on the levels that indicate a higher/lower risk of developing certain medical conditions. In the data pertaining to this report, plasma beta-carotene levels are documented in units of ng/ml (nanograms per milliliter), while the cut-off value we base our response variable on is given in μmol/l. Thus, we need to express the cut-off value 0,42 μmol/l in ng/ml. To convert between the units, we use the molar mass of beta-carotene, which is 536.9 g/mol (https://pubchem.ncbi.nlm.nih.gov/compound/Beta-Carotene). Since $1\,\mu\text{g} = 1000\,\text{ng}$ and $1\,\text{L} = 1000\,\text{mL}$, we have

$$1\,\mu\text{g/L} = 1\,\text{ng/mL}.$$

Thus, the corresponding mass concentration is

$$\begin{aligned} 0.42 \times 536.9 &= 225.498\,\mu\text{g/L} \\ &= 225.498\,\text{ng/mL} \\ &\approx 225.5\,\text{ng/mL}. \end{aligned}$$

Hence, the cut-off value is 225.5 ng/mL.

With 225.5ng/mL as cut-off, there are 235 individuals in the study with "low" beta-carotene levels, and 80 individuals with "high" beta-carotene levels. This gives a ratio of $235/315 \approx 0.746$ of individuals with low beta-carotene levels to the total number of participants in the study.

The relationship between vitamin use and low beta-carotene levels will now be studied. In Table 2 the frequency of high and low beta-carotene levels for each vitamin use category is given, together with probabilities, odds for having low beta-carotene levels, and also the odds ratios.

Vitamin Use	**High**	**Low**	**Total**	**P(Low)**	**Odds(Low)**	**Odds Ratio**
No	13	98	111	0.883	7.54	1.00
Regularly	46	76	122	0.623	1.65	0.219
Irregularly	21	61	82	0.744	2.90	0.385

TABLE 2. Probability, odds, and odds ratio for low plasma beta-carotene by vitamin use category.

We now fit a logistic regression model with the binary (Bernoulli) response variable as previously described; a 1 corresponds to low beta-carotene levels (below the cut-off value 225.5ng/mL), and a 0 corresponds to high beta-carotene levels (above 225.5ng/mL). The explanatory variable is chosen as the vitamin use of the study's participants, with categories "No", "Irregularly", and "Regularly". The reference category is chosen to be the "No" category, as it is a natural baseline for the model. We call the model *Model1D*. Model1D is thus given by

$$\text{logodds}_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2},$$

where the intercept β_0 pertains to the "No" category, β_1 pertains to the "Regularly" category, and β_2 pertains to the "Irregularly" category.
In Table 3, the parameter estimates and their profile likelihood based 95% confidence intervals, pertaining to Model1D, are given.

Vitamin use category	$\hat{\beta}$	$e^{\hat{\beta}}$	**95% CI for** $\hat{\beta}$	**95% CI for** $e^{\hat{\beta}}$
No (β_0)	2.0200	7.538	(1.480, 2.646)	(4.392, 14.105)
Regularly (β_1)	-1.5179	0.219	(-2.236, -0.858)	(0.107, 0.424)
Irregularly (β_2)	-0.9537	0.385	(-1.738, -0.203)	(0.176, 0.817)

TABLE 3. Table of estimates with pertaining 95% profile likelihood based confidence intervals for the categories of vitamin use in Model1D.

Some of the values in Table 3 we recognize from Table 2. For categories "Regularly" and "Irregularly", the odds ratios in Table 2 correspond to the odds estimates in the $e^{\hat{\beta}}$-column in Table 3. Since the "No" category is chosen as reference category, the odds ratio for this category is equal to 1. Instead, the odds for having low beta-plasma levels for the "No"-category in Table 2 corresponds to $e^{\hat{\beta}_0}$ in Table 3. The odds for categories "No", "Regularly", and "Irregularly" are given by $e^{\beta_0}, e^{\beta_0+\beta_1}$ and $e^{\beta_0+\beta_2}$, and the probabilities are given by $\frac{e^{\beta_0}}{1+e^{\beta_0}}$, $\frac{e^{\beta_0+\beta 1}}{1+e^{\beta_0+\beta_1}}$, and $\frac{e^{\beta_0+\beta_2}}{1+e^{\beta_0+\beta_2}}$, respectively.

We now calculate the linear predictor using the logistic regression model Model1D. The linear predictor, odds, and probability of having low beta-catorene levels, together with 95% Wald confidence intervals for each of the vituse categories, are shown in Table 4.

Vituse	Pred.	95% CI (Pred.)	Odds	95% CI (Odds)	Probability	95% CI (Prob.)
No	2.020	(1.441, 2.599)	7.538	(4.227, 13.444)	0.883	(0.809, 0.931)
Regularly	0.502	(0.136, 0.868)	1.652	(1.146, 2.383)	0.623	(0.534, 0.704)
Irregularly	1.066	(0.570, 1.562)	2.905	(1.769, 4.770)	0.744	(0.639, 0.827)

TABLE 4. Predicted logit values, odds, and probabilities with 95% Wald confidence intervals for each vitamin use category in Model 1D.

We see that the values in Table 4 matches those in Table 2.

We are now interested in testing whether there is any significant difference between the categories of vitamin use in Model1D. To do this, we use a partial likelihood ratio test. Since Model1D only incorporates the vitamin use variable, it is in fact the same as a global likelihood ratio test. The details for the test are given in Table 5.

Test	H_0	**Test statistic**	**Asymptotic distribution**	**p-value**	**Conclusion**
Partial likelihood test	$\beta_1 = \beta_2 = 0$	21.838	$\chi^2(2)$	$1.811 \cdot 10^{-5}$	Reject H_0

TABLE 5. Table containing test name, null hypothesis, test statistic, the test statistic's asymptotic distribution under the null hypotheses, p-value and conclusion, for Model1D.

Seeing as the p-value is less than a significance level of 0.01, we can confidently reject the null hypothesis; there is a significant difference between the categories of vitamin usage. Thus, vitamin usage does affect the probability of having low beta-carotene levels.

2. Low plasma beta-carotene and BMI

In this section, the relationship between β-carotene plasma levels and BMI was studied. This relationship was evaluated by a logistic regression model with BMI as the regressor as follows:

$$\text{logodds}_i = \beta_0 + \beta_{\text{BMI}} x_{i,\text{BMI}} \tag{2.1}$$

We denote this model as *Model.2(a)*. The same cut-off point of 225.5 ng/mL as in Section 1 was utilized.

Model.2(a)'s predicted probabilities with its 95% confidence interval, together with the data and a moving average, can be found in Figure 1. The moving average is contained in the confidence interval, which indicates that the model performs reasonably well. When BMI increases, the probability of having low β-carotene plasma levels also increases. The parameter and odds ratios of *Model.2(a)* are presented in Table 6.

The odds ratio associated with a change in BMI by an amount Δ is given by:

$$\text{OR} = \frac{\exp\left(\beta_0 + \Delta\beta_{\text{BMI}}\right)}{\exp\left(\beta_0 + \beta_{\text{BMI}}\right)} = \exp(\Delta\beta_{\text{BMI}}) \tag{2.2}$$

The odds ratios for the changes of $(+1, -1, -10)$ in BMI are found in Table 7.

TABLE 6. Parameter estimates and profile likelihood-based confidence intervals for Model 2(a)

Variable	$\hat{\beta}$	**95% CI (β)**	**OR (e^{β})**	**95% CI (OR)**
Intercept	-1.2667	[-2.7116, 0.0634]	0.2817	[0.0664, 1.0656]
BMI	0.0925	[0.0403, 0.1510]	1.0969	[1.0411, 1.1631]

TABLE 7. Odds Ratios (OR) for specified BMI changes with 95% confidence intervals.

Change in BMI	**Odds Ratio (OR)**	**Lower 95% CI**	**Upper 95% CI**
+1 unit	1.097	1.041	1.163
−1 unit	0.912	0.860	0.960
−10 units	0.397	0.221	0.668

To see whether the β_{BMI} parameter is significant, both a Wald and Likelihood Ratio test were performed. Both tests have the following null hypothesis:

$$H_0 : \beta_{\text{BMI}} = 0$$
$$H_1 : \beta_{\text{BMI}} \neq 0,$$

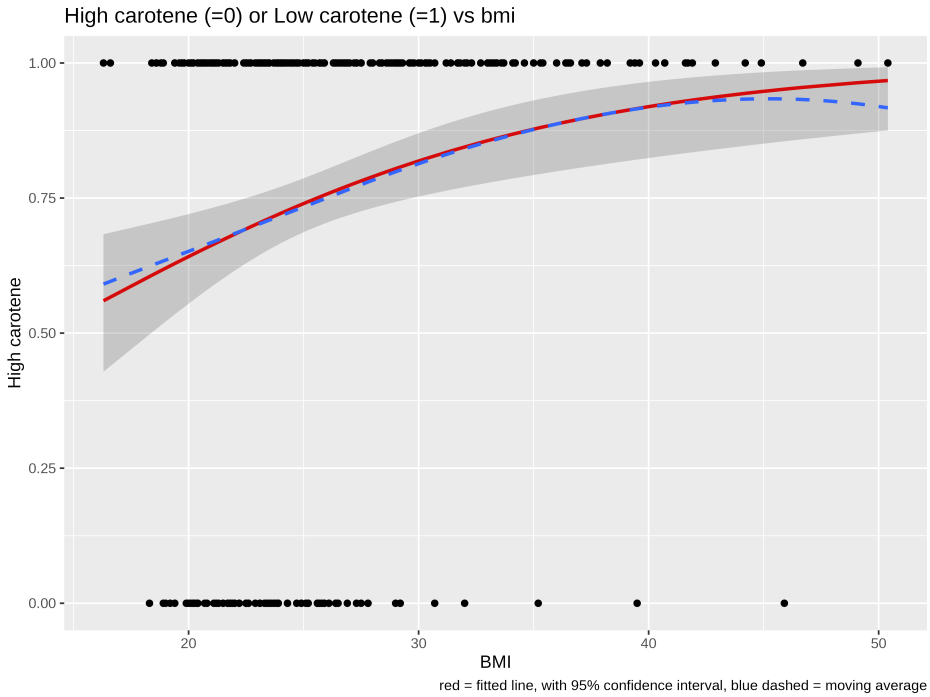


FIGURE 1. *Model.2(a)* predicted probability of having low β-carotene plasma levels, together with a moving average of the data.

but they have different test statistics. The Wald test utilizes a normal approximation of the parameter, given by:

$$z = \frac{\hat{\beta}_{\text{BMI}}}{\text{SE}(\hat{\beta}_{\text{BMI}})} \sim N(0,1) \tag{2.3}$$

Whereas the Likelihood ratio test compares a simpler model without β_{BMI} with one with the parameter and compares the log likelihood of each model, as follows:

$$LR = -2 \times [\log(L_{\text{null}}) - \log(L_{\text{full}})] \sim \chi^2(df) \tag{2.4}$$

The degrees of freedom are the number of extra parameters in the full model compared to the null. Hence, in our case, the test statistic has distribution $LR \sim \chi^2(1)$. The advantage of the Likelihood ratio test is that the test statistic converges faster to the limit distribution than the Wald test statistic, making the Likelihood ratio test more suitable for smaller data sets.

Both tests consider the β_{BMI} parameter significant, as seen in Table 8. Additionally, both tests yield very small p-values—especially the likelihood ratio test—hence we can confidently reject the null hypothesis.

To compare the leverage between our logistic model *Model.2(a)* and a linear model, the following linear model was fitted:

$$Y_i = \beta_0 + \beta_{\text{BMI}} x_{i,\text{BMI}} + \epsilon_i \tag{2.5}$$

Where Y_i is the indicator variable, low β-carotene levels is represented by $(Y = 1)$ and high by $(Y = 0)$. The leverage of both models is plotted in Figure 2. The leverage for the linear model only depends on the independent variable BMI, while for the logistic model, the leverage also depends on fitted probabilities and therefore the dependent variable. Observations further from the mean of the BMI have a high leverage for the linear model. In contrast, the logistic model has comparatively lower leverage for high BMI observations. This means that the data points with high BMI have the ability to affect the regression coefficient more for the linear model.

In Figure 3, the standardized residuals of the model are plotted. The 3 observations with the highest BMI values, which also had β-carotene values above the cutoff have standardized residuals greater than 2

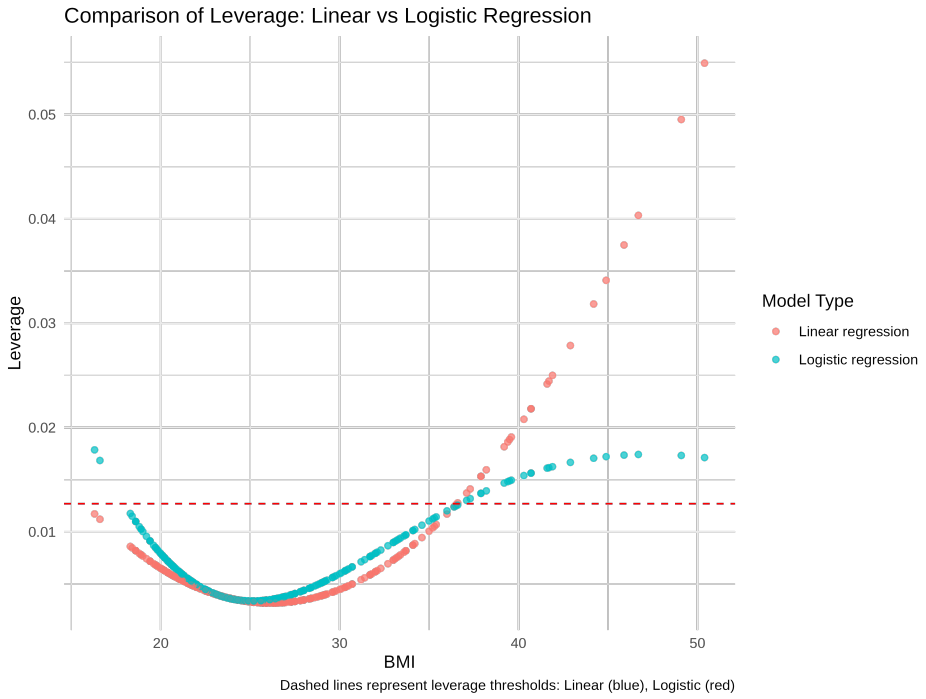


FIGURE 2. Leverage of the linear regression model and the logistic *Model.2(a)*.

standard deviations from the mean. This is expected since the 3 observations are the furthest away from the predicted probability, see Figure 1.

Due to being so far away from the predicted probability, these 3 observations also have the highest Cook's distance of all the observations, as illustrated in Figure 4. Since the predicted probability increases with BMI, the highest Cook's distance is found for the biggest BMI value.

TABLE 8. Comparison between Wald and Likelihood Ratio tests for BMI in *Model 2(a)*.

Test	**Test statistic**	**Distribution under H_0**	**P-value**
Wald test	$z = 3.283$	$z \sim N(0,1)$	0.00103
LR test	$LR = 13.196$	$\chi^2(1)$	0.0002806

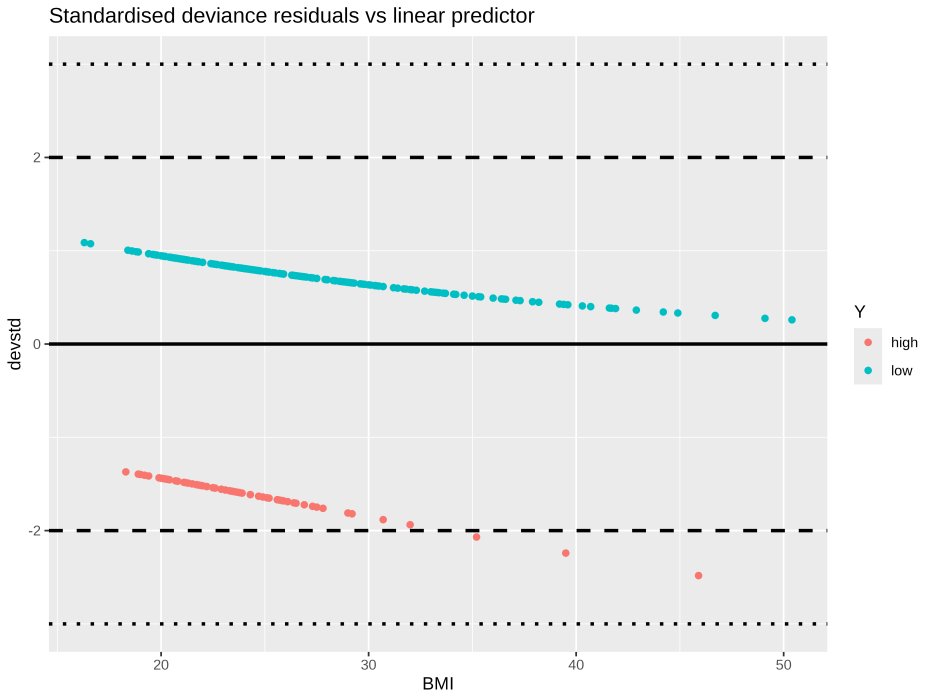


FIGURE 3. Standardized residuals of the logistic model *Model.2(a)*.

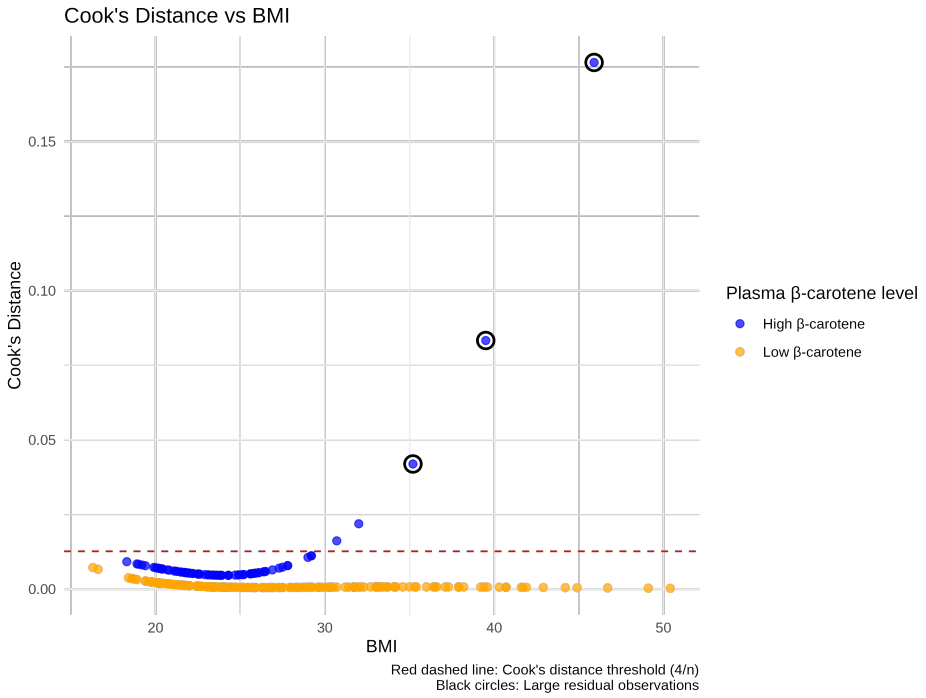


FIGURE 4. Cook's distance of the obestvations of the logistic model *Model.2(a)*.

3. MULTIPLE LOGISTIC REGRESSION AND MODEL SELECTION

We now fit a logistic regression model with the 11 covariates mentioned in the introduction in Table 1. The $\text{GVIF}^{1/2f}$ values for this model are given in Table 9.

Variable	**GVIF$^{1/2f}$**
Age	1.123312
Sex	1.128717
Smoking	1.025246
BMI	1.036662
Vitamins	1.031496
Calories	3.259374
Fat	2.696217
Fiber	1.605234
Alcohol	1.150276
Cholesterol	1.352103
Beta-carotene	1.145212

TABLE 9. Explanatory variables with pertaining GVIF$^{1/2f}$-values in a model using all 11 explanatory variables.

In Table 9, we notice that the GVIF$^{1/2f}$ for the calories category is large, indicating a high dependence between this variable and the other explanatory variables. Refitting the model without the calories variable, we get GVIF$^{1/2f}$ values for the new model, shown in Table 10.

Variable	**GVIF**
Age	1.101561
Sex	1.130991
Smoking	1.024029
BMI	1.034545
Vitamins	1.026870
Fat	1.359397
Fiber	1.172200
Alcohol	1.053184
Cholesterol	1.341352
Beta-carotene	1.140756

TABLE 10. Explanatory variables with pertaining GVIF-values in a model using every available covariate, other than the calories covariate.

We note that removing the calories category also solved the problem of a high GVIF$^{1/2f}$ for the fat category, which was seen in Table 9. Furthermore, note that since the VIF only depends on the correlation between covariates, a linear regression model with the same covariates will have the same VIFs as this logistic regression model.

We now fit a model with the variables given in Table 10 *and* their two-way interactions. We call this model the full model. Starting with the full model, we do a one-step *Backwards elimination*, and starting with the null model, we do a one-step *Forward selection*. These steps are performed with the BIC criterion. Label the resulting models the Backward model and Forward model, respectively. The variables of the models and the β estimates are given in Table 12.

With the null and full models as scope, we now perform a *Stepwise regression* on the Backward and Forward model. The variables and β-estimates for the Stepwise backward model are given in Table 12. The model obtained from doing stepwise regression on the forward model is the same as the Forward model, meaning that it was already "locally optimal"; no variables were removed. Thus, we call the model obtained from stepwise regression on the Forward model simply the Forward model; seeing as they are the same model.

We call the model obtained from stepwise regression on the Backward model the Stepwise from Backward model. The variables and β-estimates for the Stepwise from Backward model are given in Table 12.

In the Forward model, we notice that the categorical variable vituse is included. It may be possible to further reduce this model, merging the "Regularly" and "Irregularly" categories into one category. We call this reduced category "User". To see if this is reasonable, we fit a new model with the reduced variable and compare it to the larger model with a likelihood ratio test. The details for the test are given in Table 11.

Test	H_0	**Test stat.**	**Asymptotic dist.**	**p-val**	**Conclusion**
LRT	Full model = reduced model	0.35	$\chi^2(1)$	0.5541	Do not reject H_0

TABLE 11. Likelihood ratio test comparing full Forward model vs. reduced model.

Since the p-value is way larger than 0.05, we conclude that we can reduce the model to contain only two vitamin categories. We call this model the Reduced Vituse model.

In Table 12 the variables and β-estimates for the Reduced Vituse model are presented, together with the other models discussed in this section. We notice that the Backwards based models contain more variables, implying that backwards procedures might give more complex models, compared to forward procedures (of course, here we need to take into account that we only did one step in each direction). We also notice how the betadiet variable and the intercept contribute significantly less in the Backward model, owing to emitted/omitted variables.

Variable	**Backward**	**Forward**	**Stepwise Bwd**	**Reduced Vituse**
(Intercept)	0.8543	6.0612	4.4788	6.1300
bmi	0.0870	-0.0565	-0.0407	-0.0567
age	-0.0281	-0.0353	-0.0293	-0.0363
fat	-0.0554		-0.0569	
cholesterol	0.0112		0.0109	
fiber	0.1444		0.1303	
betadiet	-0.2434	-2.1308	-2.0428	-2.1606
fat:betadiet	0.0280		0.0291	
cholesterol:betadiet	-0.0045		-0.0044	
fiber:betadiet	-0.0822		-0.0745	
vituseRegularly		-1.4599		
vituseIrregularly		-1.2516		
betadiet:bmi		0.0723		0.0734
bmi:betadiet			0.0627	
vituse_reducedUser				-1.3901

TABLE 12. Estimated β-parameters from each of the fitted models: backward, forward (same as stepwise forward), stepwise backward, and reduced-vitamin use. Empty cells indicate that the variable was not included in the respective model. The variable1:variable2 notation indicates an interaction variable between variable1 and variable2.

We now want to compare the models given in Table 12, using McFadden's adjusted pseudo R^2, AIC and BIC. The values for each model are given in Table 13.

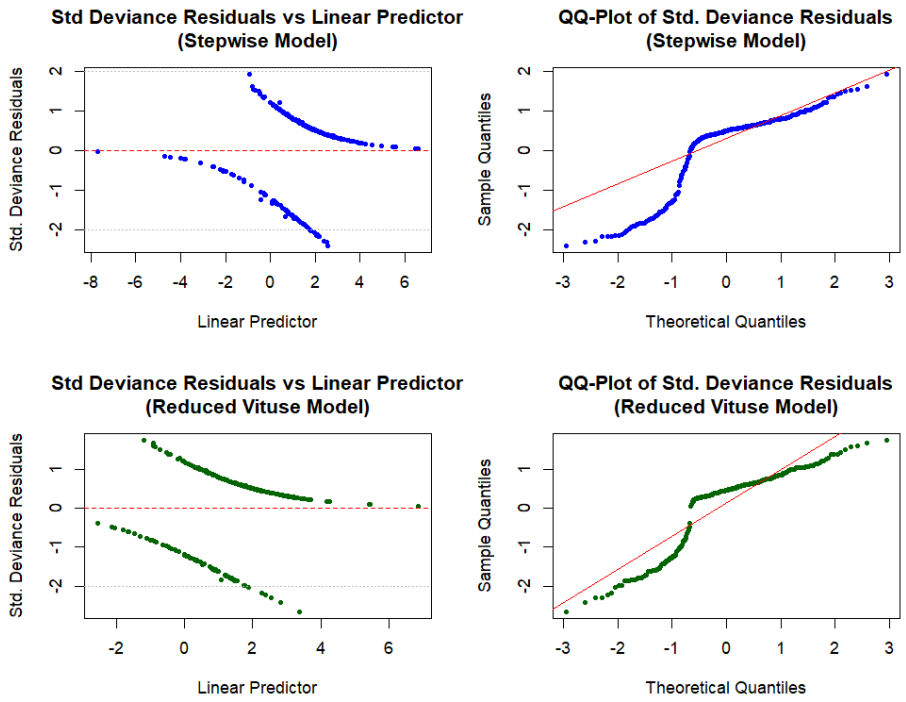


FIGURE 5. Standardized deviance residuals vs. linear predictors and QQ-plots for the Stepwise (from Backward) and Reduced Vituse models.

Model	**McFadden's R^2_{adj}**	**AIC**	**BIC**
Backward Model	0.1820	303.03	340.56
Forward Model	0.1775	301.64	327.90
Stepwise from Backward	**0.1985**	**298.12**	339.40
Reduced Vituse Model	0.1793	299.99	**322.50**

TABLE 13. Comparison of fitted models based on McFadden's adjusted pseudo R^2, AIC, and BIC. The best model, according to each metric is marked in bold.

The best models, according to McFadden's adjusted psudo R^2, AIC and BIC have been marked with bold in Table 13. Thus, the Stepwise from Backward model is favoured by McFadden's adjusted pseudo R^2 and AIC, while the Reduced Vituse model (the reduced Forward model) is favoured by BIC.

Further, we want to do a comparison between the residuals for the Stepwise from Backward model and Reduced Vituse model. The plots of the standardized deviance residuals against the linear predictors for the two models, and the QQ-plots of the models' residuals, are given in Figure 5.

In the QQ-plots, the standardized deviance residuals for the Reduced Vituse model seem to closer follow a normal distribution, as they visibly deviate less than those for the Stepwise from Backward model. In the standardized deviance vs. linear predictor plots, we also notice that the standardized deviance residuals behave more well; there are visibly fewer residuals larger than 2 (or -2) in the Reduced Vituse model compared to the Stepwise from Backward model.

4. Goodness-of-fit

In this section, we evaluate the goodness-of-fit of the *Stepwise Bwd Model* and *Red. Vituse Model* from section 3. We utilized a classification threshold of 0.5; for both models, we deemed predicted probabilities $\hat{p}_i > 0.5$ as "should have low concentration" and $\hat{p}_i \leq 0.5$ as "should not have low concentration". Using these predicted classifications and the observed concentrations of β-carotene, we constructed confusion matrices for the *Stepwise Bwd Model* and the *Red. Vituse Model*, presented in Table 14 and 15 respectively. The *Red. Vituse* model had 2 more false negatives than *Stepwise Bwd Model*; otherwise, the models performed the same.

Table 14. Confusion matrix for Stepwise from Backward model with threshold of 0.5. TN means correctly predicting high β-carotene, TP means correctly predicting low β-carotene, FP means incorrectly predicting low β-carotene, and FN means incorrectly predicting high β-carotene.

Predicted ($\hat{Y}_i$) / True (Y_i)	**High ($Y_i = 0$)**	**Low ($Y_i = 1$)**	**Total**
Failure ($\hat{p}_i \leq 0.5$)	27 (TN)	10 (FN)	37
Success ($\hat{p}_i > 0.5$)	53 (FP)	225 (TP)	278
Total	80	235	315

Table 15. Confusion matrix for Reduced Vituse model (threshold = 0.5). TN means correctly predicting high β-carotene, TP means correctly predicting low β-carotene, FP means incorrectly predicting low β-carotene, and FN means incorrectly predicting high β-carotene.

Predicted ($\hat{Y}_i$) / True (Y_i)	**High ($Y_i = 0$)**	**Low ($Y_i = 1$)**	**Total**
Failure ($\hat{p}_i \leq 0.5$)	26 (TN)	13 (FN)	39
Success ($\hat{p}_i > 0.5$)	54 (FP)	222 (TP)	276
Total	80	235	315

The classification performance of both models is summarized in Table 16. The *Stepwise Bwd Model* performs slightly better and is notably the only one of the two models that passes the Accuracy vs. No Information Rate (NIR) test, which tests whether the model has better accuracy than a model that always predicts the most common group (low β-carotene levels). Similarly, the McNemar's test compares the model's disagreement between predicted and observed classification marginals; both our models pass this test.

Table 16. Predictive performance comparison at 0.5 threshold for the *Stepwise Bwd Model* and the *Red. Vituse Model*.

Measure	**Stepwise from Backward**	**Reduced Vituse**
Accuracy	0.8000	0.7873
P-value (Accuracy > NIR)	0.0147	0.0508
Cohen's κ	0.3585	0.3245
McNemar's Test (P-value)	<0.001	<0.001
Sensitivity	0.9574	0.9447
Specificity	0.3375	0.3250

The Receiver Operating Characteristic curves (ROC) show the possible sensitivity and specificity performance of the models for different classification thresholds, and this curve is plotted in Figure 6. Additionally,

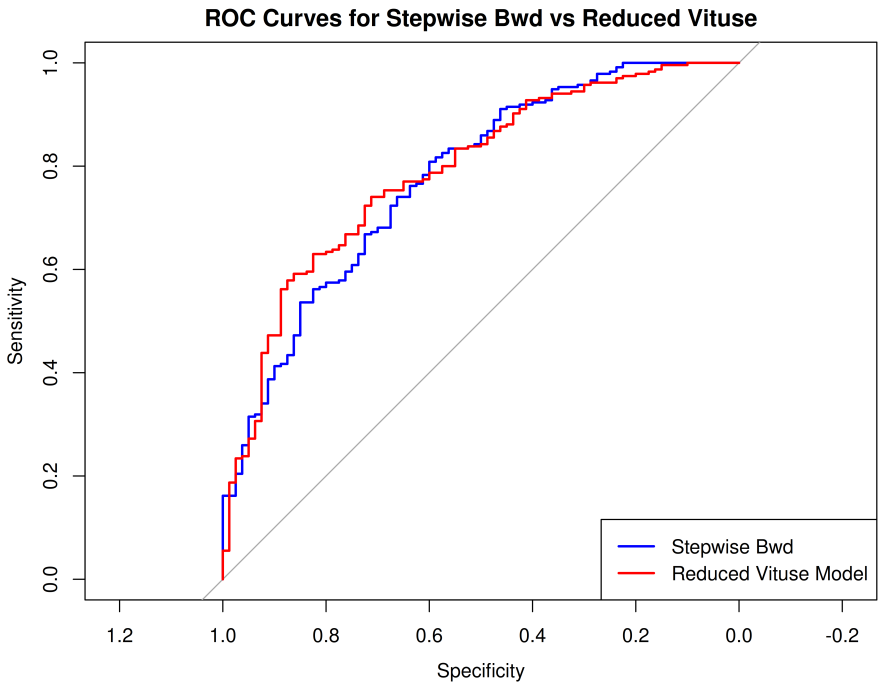


FIGURE 6. ROC curves of the *Stepwise Bwd model* and *Reduced Vituse model*.

the area under the curve (AUC) is presented in Table 17.The Reduced Vitamin Use Model has a marginally higher AUC (0.7883 compared with 0.7874), but the difference is negligible. The paired DeLong test provides no evidence that the models differ in their ability to discriminate between low and high plasma β-carotene concentrations.

TABLE 17. AUC values with 95% confidence intervals for the Stepwise Backward and Reduced Vituse models.

Model	**AUC**	**Lower 95% CI**	**Upper 95% CI**
Stepwise Backward	0.7874	0.7285	0.8464
Reduced Vituse	0.7883	0.7316	0.8450

Utilizing the ROC curves an optimal threshold was found by minimizing:

$$(1 - \text{Sensitivity})^2 + (1 - \text{Specificity})^2$$

These new thresholds—together with the corresponding sensitivity and specificity—can be found in Table 18. With the optimal thresholds, the models have identical sensitivity, but the *Red. Vituse Model* have better specificity.

From these new optimal thresholds, new confusion matrices were constructed, see Table 19 and 20. With the new optimal thresholds, the *Red. Vituse Model* performed the best by having fewer false positives; otherwise, the models performed identically. By choosing the optimal threshold values—instead of 0.5—both models' specificity increased significantly, from 0.3375 to 0.7125 for the *Stepwise Bwd Model* and 0.3250 to 0.7125 for the *Red. Vituse model*. The models make significantly fewer false positives, but as a tradeoff, the sensitivity of the models have decreased, from 0.9574 to 0.7149 for the *Stepwise Bwd Model* and 0.9447 to 0.7404 *Red. Vituse Model*. Additionally, the accuracy of the models has decreased to such a degree that none of them pass the Accuracy vs. No Information Rate test. Cohen's κ has increased for both models when using the optimal threshold. Cohen's κ measures how much better the model's classification performs than a model that randomly classifies an observation based on the proportion of each group.

Overall, the best model is the *Red. Vituse model* with the optimal threshold. The odds ratios and corresponding confidence intervals are presented in Table 22. The confidence interval of the odds ratio for the BMI contains 1, suggesting the effect could be insignificant. Additionally, the odds ratio of the age ($OR = 0.96$) indicates a weak effect that age lowers the risk of having low β-carotene levels. Maybe age-related reduction in metabolism decreases how quickly β-carotene is converted to vitamin A in the body, and therefore higher β-carotene levels in the blood are measured. The interaction effect between BMI and β-carotene intake could be that overweight reduces the body's ability to absorb β-carotene.

Unsurprisingly, dietary intake of β-carotene strongly decreases the risk of having low β-carotene levels. Almost as strong of an effect is seen from vitamin use, likely because a lot of vitamin supplements contain β-carotene.

TABLE 18. Optimal thresholds and their corresponding sensitivity and specificity.

Model	**Optimal threshold**	**Sensitivity**	**Specificity**
Stepwise from Backward	0.7795	0.7149	0.7125
Reduced Vituse	0.7340	0.7404	0.7125

TABLE 19. Confusion matrix for Stepwise from Backward model (optimal threshold = 0.7795). TN is the true negative, TP is the true positive, FP is the false positive, and FN is the false negative.

Predicted ($\hat{Y}_i$) / True (Y_i)	**High ($Y_i = 0$)**	**Low ($Y_i = 1$)**	**Total**
Failure ($\hat{p}_i \leq 0.7525$)	57 (TN)	67 (FN)	124
Success ($\hat{p}_i > 0.7525$)	23 (FP)	168 (TP)	191
Total	80	235	315

TABLE 20. Confusion matrix for Reduced Vituse model (optimal threshold = 0.7340). TN is the true negative, TP is the true positive, FP is the false positive, and FN is the false negative.

Predicted ($\hat{Y}_i$) / True (Y_i)	**High ($Y_i = 0$)**	**Low ($Y_i = 1$)**	**Total**
Failure ($\hat{p}_i \leq 0.7340$)	57 (TN)	61 (FN)	118
Success ($\hat{p}_i > 0.7340$)	23 (FP)	174 (TP)	197
Total	80	235	315

TABLE 21. Predictive performance comparison at optimal thresholds

Measure	**Stepwise from Backward**	**Reduced Vituse**
Accuracy	0.7143	0.7333
P-value (Accuracy > NIR)	0.9116	0.7222
Cohen's κ	0.3618	0.3916
McNemar's Test (P-value)	<0.001	<0.001
Sensitivity	0.7149	0.7404
Specificity	0.7125	0.7125

TABLE 22. Odds ratios and their 95% profile likelihood confidence intervals for the Reduced Vituse Model.

Parameter	**Odds Ratio**	**Lower 95% CI**	**Upper 95% CI**
Intercept	459.43	20.96	10485.83
BMI	0.945	0.850	1.057
Age	0.964	0.945	0.983
Betadiet	0.115	0.030	0.377
Vitamin use (reduced)	0.249	0.119	0.488
BMI:Betadiet	1.076	1.026	1.136

5. AI AND AUTHOR CONTRIBUTIONS

AI was used to generate tables, and in some instances, to improve the code that generated the plots. A substantial amount of the code utilized in the analysis was inspired from the lectures.

Christian Bjur has written Sections 2 and 4. Similarly, Leonard Nygren Löhndorf has been responsible for Sections 1 and 3. Both authors have proofread the work.